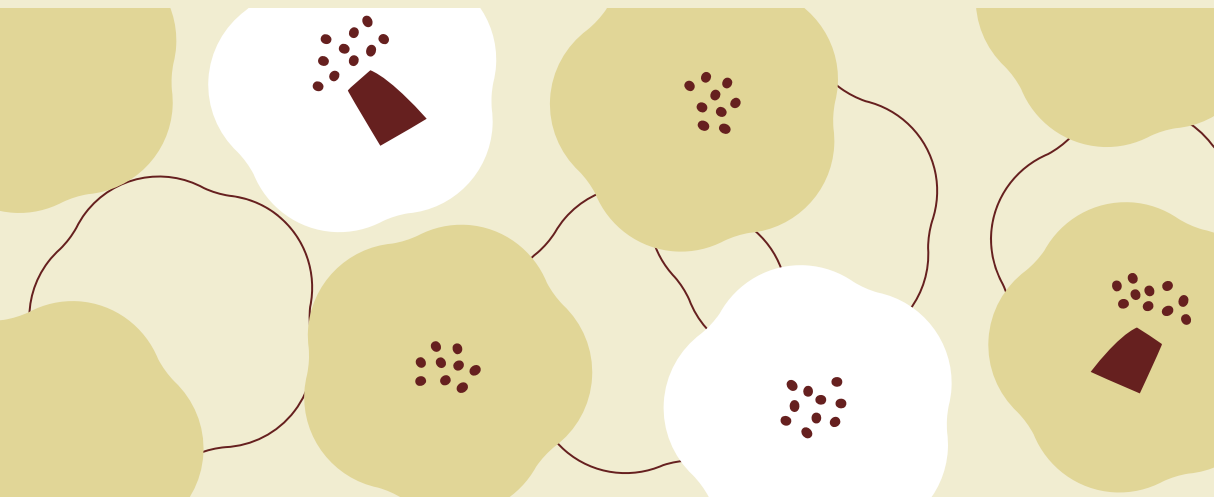




CS-510 (Information and Randomness)

Part-1 (Lectures)

Part-2 (Seminar and Term paper) - (10 + 10 = 20%)

Midsem = 40%

Endsem = 40%

Books :-

- 1) Introduction to Cryptography with Coding Theory - Trappe and Washington
- 2) Handbook of Applied Cryptography - Menzen, Van Oorschot, Vamhorne.
- 3) Introduction to Cryptography - Stinson

Lecture-1 (25/7/25)

Pseudo Random Number Generation

(HW)

- 1) Elapsed time between emission of particles during radioactive decay.
- 2) Thermal noise from a semiconductor diode or register
- 3) Frequency instability of a free running oscillator.
- 4) The amount of a metal insulator semiconductor capacitor is charged during a fixed period.
- 5) Air turbulence within sealed disk drive which causes random fluctuation in disk drive sector read latency time.
- 6) Sound from microphone or video input from a camera.

Software Based Generators

- 1) The system clock
- 2) Elapsed time between key strokes or mouse movement.
- 3) Content of input/output buffer
- 4) O/S system values such as system load and network statistics.

Deskewing

Output bits may be biased (the probability of a source emitting $1 \neq \frac{1}{2}$) or correlated (the prob of a source emitting depends on the previous bit)

Deskewing :- Removing biases in output bits

$$Pr(X=1) = p \quad 0 < p < 1$$

$$Pr(X=0) = (1-p)$$

Output sequence of such generators

Pairs		Prob
00	X	$(1-p)^2$
01	0	$(1-p)p$
10	1	$p(1-p)$
11	X	p^2

Defⁿ 1:- A random bit generator is a device or algorithm whose output sequence is abstractly independent and unbiased binary digits.

Defⁿ 2:- A (PRBG) pseudo random bit generator is a deterministic algorithm which given a truly random binary sequence of length K , outputs a binary sequence of length $L \gg K$ which appears to be random. The input to PRBG is the seed, which out of PRBG is pseudo random bit generator.

rand() function is a linear congruential generator. It produces a pseudo random sequence $x_1, x_2, x_3, \dots$ according to a linear recurrence.

$$x_n = (ax_{n-1} + b) \bmod m.$$

Here a, b and m are parameters which characterise the generator and x_0 is the initial seed.

This is not cryptographically secure and fails many statistical tests.

Method for pseudo random bit generator:-

One Way Functions :-

$f()$ is one way function if given x , it is computationally feasible to compute $y = f(x)$, but given y it is computationally infeasible to get x .

Define:-

→ One way function

$$x_j = f(s+j) \text{ for } j=1, 2, 3, \dots$$

If we let b_j be the least significant bit of x_j then the sequence of b_j s is generated pseudo random number.

L-2

Defⁿ :- A pseudo random bit generation is said to pass all polynomial time statistical tests if no polynomial time algorithm can correctly distinguish between the output sequence of the generator and truly random sequence of the same length with probability significantly greater than $1/2$.

Defⁿ: A pseudo random bit generation is said to pass next bit test, if there is no polynomial time algorithm which on input of the first l bits of an output sequence 's', can predict $(l+1)^{st}$ bit of 's' with probability greater than $\frac{1}{2}$ (significantly greater)

Fact :- (universality of the next bit test)

→ A pseudo random bit generator passes the next bit test if and only if it passes all the polynomial time statistical tests.

Defⁿ:- PRBG that passes next bit tests (possibly under some plausible but unproven mathematical assumptions such as the intractability of factoring integers) is called cryptographically secure PRBG.

Colomb's Randomness Postulate

Defⁿ:- Let $S = s_0 s_1 s_2$ be an infinite sequence. The subsequence consisting of the first n terms of s is denoted by $S^n = s_0, s_1, \dots, s_{n-1}$

Defⁿ:- The seq $S = s_0, s_1, s_2, \dots$ is said to be N periodic, if $S_i = S_{i+N} \forall i \geq 0$. The seq s is periodic if it is N -periodic for some +ve integer N . The period of the sequence s , is the smallest positive integer N which is N -periodic. If s is periodic sequence of period N , then the cycle of s is the subsequence of S^N .

Defⁿ:- Let s be a sequence. A run of s is a subsequence of ' s ' consisting of consecutive 0's or consecutive 1's which is neither preceded or succeeded by the same symbol. A run of 0's is called a gap and a run of 1's is called block.

Defⁿ: Let $s = s_0 s_1 s_2 \dots$ be a periodic sequence of period N . The auto correlation function of s is the integer valued function $C(t)$ defined as

$$C(t) = \frac{1}{N} \sum_{i=0}^{N-1} (2s_i - 1)(2s_{i+t} - 1)$$

for $0 \leq t \leq N-1$.

The auto-correlation function $c(t)$ measures the amount of similarity between the sequence s and a shift of s by t positions. If s is a random periodic seq of period ' N ' then $|N \cdot C(t)|$ can be expected to be quite small $\forall t$ $0 < t < N$.

Defⁿ: Let s be a periodic sequence of period N .
 Colomb's randomness postulates:-

R1: In the cycle s^N of s the number of 1's and the number of 0's differ by at most 1.

R2: In the cycle s^N atleast $\frac{1}{2}$ of the runs have length 1, $\frac{1}{4}$ of them have length 2, at least $\frac{1}{8}$ have length 3, etc as long as number of runs so indicated exceeds 1. Moreover for each

of these lengths there are (almost) enough many gaps and blocks.

R3: The auto correlation function is 2-valued that is for some integers k .

$$N \cdot C(k) = \sum_{i=0}^{N-1} (2S_i - 1)(2S_{i+k} - 1) = \begin{cases} N & \text{if } k=0 \\ k & \text{if } 1 \leq k \leq N-1 \end{cases}$$

Defⁿ: A binary sequence satisfying, Golomb's postulates is called as pseudo-noise sequence or pn-sequence.

Pn-sequence

Consider the Periodic sequence

$S^{15} = \underline{0}, \underline{1}, \underline{1}, \underline{0}, \underline{0}, \underline{1}, \underline{0}, \underline{0}, \underline{0}, \underline{1}, \underline{1}, \underline{1}, \underline{1}, \underline{0}, \underline{1}$

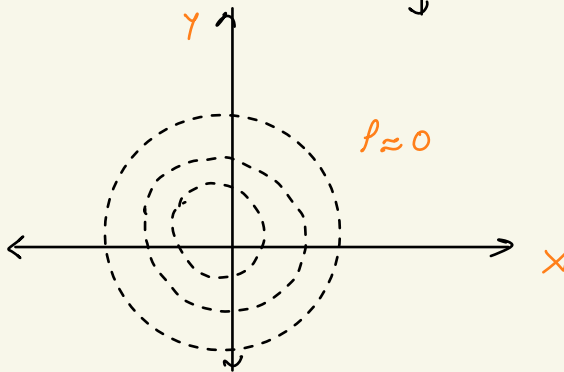
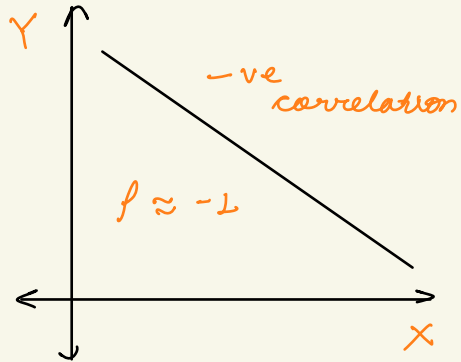
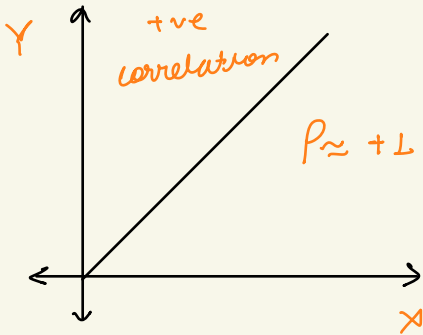
R1: # 0's = 7, # 1's = 8

R3: S^{15} has 8 runs.

There are 4 runs of length 1 (2 gaps, 2 blocks).
 There are 2 runs of length 2 (1 gap, 1 block)
 There is 1 run of length 3 (2 gap)
 There is 1 run of length 4 (1 block)

R3: The auto correlation function $C(k)$ takes on 2 values $C(0)=1$ and $C(k)=-\frac{1}{15}$ for $1 \leq k \leq 14$

$$\rho(k) = \frac{1}{N} \sum_{i=0}^{N-k} (2s_i - 1)(2s_{i+k} - 1) \quad \text{for } 0 \leq k \leq N-1$$



$\begin{matrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{matrix}$

mismatch
 $k=1$

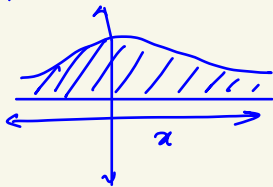
Statistical tools

Defⁿ: If the result is X of an experiment can be a real number, then X is said to be a continuous random variable.

Defⁿ: A probability density function of a continuous random variable X is a function $f(x)$ which can be integrated and satisfies.

(i) $f(x) \geq 0 \quad \forall x \in \mathbb{R}$

(ii) $\int_{-\infty}^{+\infty} f(x) dx = 1$



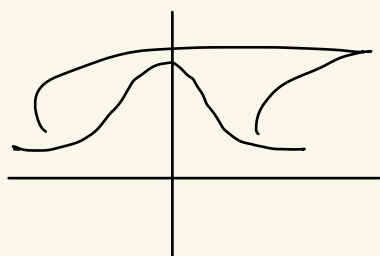
(iii) $\forall a, b \in \mathbb{R}$

$$P(a < X \leq b) = \int_a^b f(x) dx$$

Defⁿ: A continuous random variable X has a normal distribution with mean μ and variance σ^2 if its probability density function (pdf) is defined by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} \quad -\infty < x < \infty$$

X is said to be $N(\mu, \sigma^2)$. If X is $N(0, 1)$ then X is said to have standard normal distribution.



Point of
inflection

$$P(X > 1.6449) = 0.05$$

α	0.1	0.05	0.025	0.01	0.005	0.0025	0.001	0.0005
z	1.2816	1.6449	1.9600	2.3263	2.5768	2.8070	3.0902	3.2905

selected percentiles of $N(0, 1)$ if X is a random variable having standard normal distribution, the $P(X > z) = \alpha$

χ^2 distribution

→ Can be used to compare the goodness of fit of the observed frequency of events to other expected frequencies under a hypothesized distribution. The χ^2 distribution with v degrees of freedom arises in practice when the squares of v independent random variables having standard normal distributions are summed

$$X_1, X_2, \dots, X_v \in N(0,1)$$

$X_1^2 + X_2^2 + \dots + X_v^2$ in chi square (χ^2) distribution

Defⁿ:- $v \geq 1$ be an integer. A continuous X has χ^2 distribution, with v degree of freedom if pdf is given by

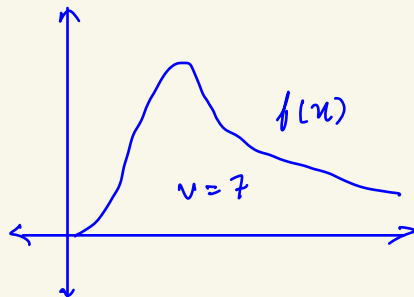
$$f(x) = \begin{cases} \frac{1}{\Gamma(v/2) 2^{v/2}} x^{(v/2)-1} e^{-x/2} & 0 < x < \infty \\ 0 & x < 0 \end{cases}$$

Γ : Gamma Function

$$\Gamma(t) = \int_0^{\infty} x^{t-1} e^{-x} \cdot dx, \text{ for } t > 0$$

The mean and variance of χ^2 -distribution are

$$\mu = v, \quad \sigma^2 = 2v$$



Hypothesis Testing

A statistical hypothesis H_0 is an assertion about a distribution of one or more random variables.

A test of a statistical hypothesis is a procedure based upon values of random variables that leads to acceptance/rejection of the hypothesis H_0 .

Defⁿ: The significance level α of the test of a statistical hypothesis H_0 is probability of rejecting H_0 when it is true.

→ Let H_0 be the hypothesis of a given binary sequence produced by random bit generator.

→ If significance level α of a test of H_0 is too high then the test may reject the sequence produced by the random bit generator (Type I error)

→ If α of the test of H_0 is too low then the test may accept sequence that are not random.
(Type II error)

→ A statistical test is implemented for specifying a statistic on a random example.

→ A statistic is a function of the elements of a random sample.

for eg. # of 0's in a binary sequence.

1) Suppose the statistic X of the random sample follow χ^2 distribution with v degree of freedom. To achieve a significance level of α , a threshold value χ_α is chosen so that $P(X > \chi_\alpha) = \alpha$

If the value X_s of the statistic of the sample output sequence satisfies $X_s > \chi_\alpha$, then the seq falls the test otherwise it passes the test. Such a test is called 1 sided test.

2) Suppose the statistic X of a random sequence follows $N(0,1)$ distribution, and suppose the statistic is expected to take on both larger and smaller values of non-random seq. To achieve significance level of α a threshold χ_α is chosen such that

$$P(X > \chi_\alpha) = P(X < -\chi_\alpha) = \alpha/2$$

If the value X_s of the statistic of the sample output seq satisfies $X_s > \chi_\alpha$ or $X_s < -\chi_\alpha$ then the seq falls the test otherwise it passes the test such a test is called 2 sided test.

5 basic tests of randomness

1) **Freq Tests** : The purpose of this test is to determine either the number of 0's and 1's in S are approximately same. Let n_0, n_1 be the no of 0's and 1's respectively. The statistic used is $\frac{(n_0 - n_1)^2}{n} = \chi^2$, χ follows χ^2 -distribution with 1-degree of freedom.

2) **Serial Test / (2-bit test)** : The purpose of this test is to determine whether the number of occurrences of 00, 01, 10 and 11 are approximately same.

$\begin{matrix} \text{00} & \text{01} & \text{10} & \text{11} \\ \downarrow & \downarrow & \downarrow & \downarrow \\ n_{00} & n_{01} & n_{10} & n_{11} \end{matrix}$

$$n_{00} + n_{01} + n_{10} + n_{11} = n - 1.$$

because subsequences are allowed to overlap.

$$\chi^2 = \frac{4}{n-1} (n_{00}^2 + n_{01}^2 + n_{10}^2 + n_{11}^2) - \frac{2}{n} (n_{00}^2 + n_{11}^2) + 1$$

will approximately follow

χ^2 distribution, with 2 degree of freedom.

3) Poker Test : Let m be a positive integer. such that $\lfloor \frac{n}{m} \rfloor \geq 5 \cdot (2^m)$ and let $k = \lfloor \frac{n}{m} \rfloor$. Divide the sequence s to k non-overlapping parts each of length m . Let n_i be the number of occurrences of i th type of seq of length m , $1 \leq i \leq 2^m$. The poker test determines whether the sequence of length m each appears same number of times in S . The statistic used is

$$\chi_3 = \frac{2^m}{k} \left(\sum_{i=1}^{2^m} n_i^2 \right) - k$$

which approximately follow χ^2 distribution with $2^m - 1$ degree of freedom.

4) Run Test : The expected number of GAP (or BLOCKS) of length i in a random seq of length

$$n \text{ is } e_i = \frac{n - i + 3}{2^{i+2}}$$

$B_i \rightarrow \# \text{ of block}$

$G_i \rightarrow \# \text{ of gaps}$

The statistic used is $\chi_4 = \sum_{i=1}^k \frac{(B_i - e_i)^2}{e_i}$

follow χ^2 distribution with $2k - 2$ degree of freedom.

Auto Correlation Test :- The purpose of this test is to check for correlations between the sequence S and (non-cyclic) shifted versions of it. Let d be a fixed integer, $1 \leq d \leq \lfloor \frac{n}{2} \rfloor$. The number of bits in S not equal to their d shifts is

$$A(d) = \sum_{i=0}^{n-d-1} s_i \oplus s_{i+d}$$

The statistic used is

$$X_5 = 2 \left(A(d) - \frac{n-d}{2} \right) / \sqrt{n-d}$$

which approx follows $N(0,1)$ if $n-d \geq 10$. This is a 2-sided test.

E.g.:- consider the sequence of length $n=160$ obtained by replicating

11100 01100 01000 10100 11101 11100 10010 01001

(i) Freq Test : $n_0=84$, $n_1=76$, $X_1=0.4$

(ii) Serial Test : $n_{00}=44$, $n_{01}=40$, $n_{10}=40$, $n_{11}=35$, $X_2=0.6252$

(iii) Poker Test : Here $m=3$, $k=53$,

The parts 000, 001, 010, 011, 100, 101, 110, 111 appear

5, 10, 6, 4, 12, 3, 6 and 7 time $X_3=9.6415$

(iv) Runs Test : $e_1=20.25$, $e_2=10.0625$, $e_3=5$, and $k=3$.

There are 25, 4, 5 blocks of length 1, 2, 3 and 8, 20, 12

gaps of length 1, 2, 3 resp, $X_4=31.7913$

(v) Autocorrelation Test : If $d=8$, $A(8)=100$, $X_5=3.8933$. For significance level $\alpha=0.05$, the threshold values of X_1, X_2, X_3, X_4 , and X_5 are 3.8415, 5.9915, 14.0671, 9.4877, 1.96 resp.

FIPS 140-1 Statistical Tests for 20,000 bits

- 1) Monobit Test : # of 1's should satisfy $9654 < n_1 < 10346$
- 2) Poker Test : Statistic X_3 is computed for $m=4$, The poker test is passed if $1.03 < X_3 < 57.4$
- 3) Runs Test : This is passed if 12 counts $B_i, G_i, 1 \leq i \leq 6$ are within corresponding specified interval.

Length of Runs	Required Intervals
1	2267 - 2733
2	1079 - 1421
3	502 - 748
4	223 - 402
5	40 - 223
6	90 - 223

- 4) Long Run Test : This is passed if there is no run of length 3, 4 or more

Maurer Universal Statistical Tests

- The basic idea behind the test is that it should not be possible to significantly compress (without loss of info) the output sequence, of a random bit generator.
- Thus if a sample output requires sequence of a bit generator can be significantly compressed, the generator is rejected as defective

Drawback: Frequency much longer output $X_3 = 9.915$ output seq.

The seq s is partitioned into non-overlapping L -bit blocks with left over bits discarded and no. of blocks is $Q+K$.

For each i , $1 \leq i \leq Q+K$, let b_i be the integer whose binary representation is i 's block. The blocks are scanned in order.

A table T is maintained so that $T[i]$ is the portion of the last occurrence of the block corresponding to integer i ,

$0 \leq i \leq 2^L - 1$. First Q blocks of s are used to initialise the table.

T . Q should be chosen to be at least $10 \cdot 2^L$ in order to have high likelihood that each of 2^L L -bit blocks occur at least one in the first Q blocks. The remaining K blocks are used to define the statistic X_u as follows. For each i , $Q+1 \leq i \leq Q+K$.

Let $A_i = i - T[b_i]$. $A[i]$ is since the last occurrence of block b_i . Then

$$X_u = \frac{1}{K} \sum_{i=Q+1}^{Q+K} \log A_i$$

K should be at least $1000 \cdot 2^L$ (have sample seq s must be at least $(1010 \cdot 2^L)$ bits).

Maurer's Universal Randomness Test

→ Computing the statistic X_u

INPUT: A binary sequence $S = S_0 S_1 \dots S_{n-1}$ of length n , and parameters L, Q, K .

OUTPUT: The value of statistic X_u for the sequence S .

1) Zero the table T . For j from 0 to $2^L - 1$ do the following:

$$T[j] \leftarrow 0$$

2) Initialise the table T . For i from 1 to Q do the following:

$$T[b_i] \leftarrow i$$

3) $\text{sum} \leftarrow 0$

4) For i from $Q+1$ to $Q+K$ do the following.

$$4.1 \text{ sum} \leftarrow \text{sum} + \log(i - T[b_i])$$

$$4.2 \text{ } T[b_i] \leftarrow i$$

5) $X_u \leftarrow \text{sum} / K$

6) Return X_u .

Maurer's universal statistical test. uses the computed value of X_u for the multiple output sequence S in the manner prescribed in FACT. To test the sequence S , a 2-sided test should be used with a significance level α between 0.001 and 0.01.

FACT: Let X_u be the statistic defined as

$$X_u = \frac{1}{K} \sum_{i=Q+1}^{Q+K} \log A_i$$

$[A_i = i - T[b_i]]$ having mean μ and variance σ^2 as given in the table. Then for random sequence, the statistical

$$Z_u = \frac{X_u - \mu}{\sigma}$$

approx follows $N(0,1)$ distribution.

L	μ	σ^2
1	0.7326496	0.690
2	1.5374387	1.335
	$\vdots$	
16	15.167379	3.421

Table: Mean μ and variance σ^2 of the statistic X_n for random sequences with parameters L, k as $Q \rightarrow \alpha$. The variance of X_n is $\sigma^2 = C(L, k)$ where

$$C(L, k) \approx 0.7 - (0.8/L) + (1.6 + (12.8/L)) * k^{-4/L}$$

for $k \geq 2^L$.

#Cryptographically Secure Pseudo Random Bit Gen

→ Computationally difficult problems

- ↳ Factoring Integers
 - Classical: no known Poly($\log n$)
 - Quantum: $\exists$ Poly algo
- ↳ Discrete Logarithm Prob (DLP)

DLP:

Take Prime p , $\mathbb{Z}_p = \{0, 1, 2, \dots, p-1\}$

Addition Modulo 5 table

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

(Addition modulo 5)

$\times_5$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

(Multiplication modulo 5)

($\mathbb{Z}_5, +_5, *_5$)

$\{2, 4, 3, 1\}$ $\{2, 4, 2, 1\}$ $\{4, 1\}$

Prime p Unknown
Generate g^x modulo p ($g^x \bmod p$)

RSA pseudorandom bit generation

Summary: A pseudorandom bit sequence $z_1, z_2, \dots, z_l$ of length l is generated.

1) setup - Generate 2 secret RSA like primes p and q and compute $n = pq$ and $\phi(n) = (p-1)(q-1)$

$$n = 12$$

$$\gcd(a, n) = 1$$

Numbers < 12 and relatively prime with respect to 12 are

$$|\{1, 5, 7, 11\}| = 4$$

select a random integer e , st, $1 < e < \phi$ and $\gcd(e, \phi) = 1$.

2) select a random integer x_0 (the seed) in the interval $[1, n-1]$.

3) for i from 2 to l do the following:

3.1 $x_i \leftarrow x_{i-1}$

3.2 $z_i \leftarrow$ the least significant bit of x_i

4) Output the sequence $z_1, z_2, \dots, z_l$

Indistinguishability of Probability

Distribution:

2 properties of PRBGs:

(i) It should be fast.

(ii) It should be secure.

→ A string of l bits produced by PRBGs should look random. That is it should be impossible in an amount of time that is polynomial in k (equivalently polynomial in l) to distinguish a string of l bits produced by PRBG from a string of l truly random bits.

Let a bit generator produces "1"s with probability $2/3$ then it should be easy to distinguish the bit stream from a truly random bit stream.

Distinguishing strategy:

Given a bit string of length l denote the number of 1's in the bit string of l_1 .

On an average a random bit string of length l will contain $l/2$ 1's.

→ Our generator's bit strings expected number of 1's: $2l/3$

$$\text{Let } l_{\text{avg}} = \frac{l/2 + 2l/3}{2} = \frac{7l}{12}$$

If $l_1 > \frac{7l}{12}$ ($= l_{\text{avg}}$) we would guess the bit string was more likely to be generated from the biased generator.

$$Z^i = (z_1, z_2, \dots, z_i) \text{ } i\text{-tuple}$$

Defⁿ: Suppose P_0 and P_1 are two probability distributions on $(Z_2)^l$ of all bit strings of length l

For $j=0, 1$ and $Z^l \in (Z_2)^l$.

$P_j(Z^l)$ denote the probability of the string Z^l occurring in distribution P_j

Let $\text{dst} : (\mathbb{Z}_2)^l \rightarrow \{0, 1\}$ be a function and let $\epsilon > 0$. For

$j=0,1$. defined
$$E_{\text{dst}}(P_j) = \sum_{\{z^l \in (\mathbb{Z}_2)^l : \text{dst}(z^l)=1\}} P_j(z^l)$$

We say a distinguisher D distinguishes P_0 and P_1 provided:

$$|E_{\text{dst}}(P_0) - E_{\text{dst}}(P_1)| \geq \epsilon$$

and say that P_0 and P_1 are ϵ -distinguishers if there is an ϵ distinguisher of P_0 and P_1 .

→ Given a tuple $z^l = (z_1, z_2, \dots, z_l)$ a distinguisher might guess 0 with some probability p (depending on z^l) and hence guess 1 with probability $1-p$. In the case of randomised distinguisher

$$E_{\text{dst}}(P_j) = \sum_{z^l \in (\mathbb{Z}_2)^l} (P_j(z^l) \times p_j[\text{dst}(z^l)=1])$$

Relevance of distinguisher to PRBGs

→ Consider the seq of l bits produced by a bit generator. There are 2^l possible sequences and if l bits are chosen independently and uniformly at random, then each of the 2^l sequences will occur with probability $\frac{1}{2^l}$.

Let the uniform probability distribution be P_u . Now, consider sequences produced by bit generator f . Suppose a k -bit seed is chosen uniformly at random and generator computes bit string of length l . Then we obtain a probability distribution on the set of all bit strings of length l and denote it by P_f .

For 2^l possible sequences 2^k sequences each occur with prob $1/2^k$ and remaining $2^l - 2^k$ sequences never occur. Hence, prob. distribution P_f is very non-uniform.

E.g.: suppose that a bit generator f only produces sequences in which exactly $l/2$ bits have the value 0 and $l/2$ bits have value 1. Define

$$\text{dist}(z_1, z_2, \dots, z_l) = \begin{cases} 1 & \text{if } (z_1, z_2, \dots, z_l) \\ & \text{has } l/2 \text{ bits of 0.} \\ 0 & \text{otherwise} \end{cases}$$

$$E_{\text{dist}}(P_u) = \frac{\binom{l}{l/2}}{2^l} \quad \text{As } \lim_{l \rightarrow \infty} \frac{\binom{l}{l/2}}{2^l} = 0$$

$E_{\text{dist}}(P_f) = 1$ for any defined value of $\epsilon < 1$.
 P_u and P_f are ϵ -distinguished.

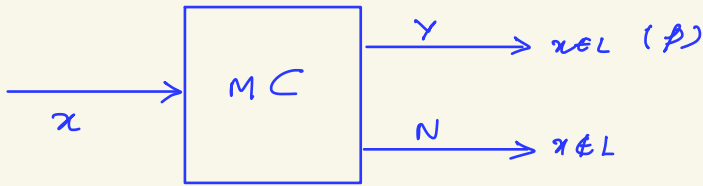
Next Bit Predictor

→ Let f be a (k, l) bit generator. Suppose $1 \leq i \leq l-1$, we have the function

$$nbp: (\mathbb{Z}_2)^{i-1} \rightarrow \mathbb{Z}_2$$

which takes input (-2) tuple $z^{i-1} = (z_1, z_2, z_3, \dots, z_{i-1})$

Security of BBS Generator



Theorem: Suppose A is an unbiased Monte Carlo Algorithm with error probability at most $\frac{1}{2} - \delta$. Suppose, we define an algorithm A^n by running A , $n = 2m + 1$ times on a given instance I and outputting the most frequent answer. Then the error probability of algorithm A^n is at most

$$\frac{(1 - 4\delta^2)^m}{2}$$

Pf: The probability of obtaining exactly i correct answer in n trials is at most:

$$\binom{n}{i} \left(\frac{1}{2} + \delta \right)^i \left(\frac{1}{2} - \delta \right)^{n-i}$$

The probability that most frequently occurring answer is incorrect is equal to the probability that the number of correct answer in n trials of almost m .

$$\begin{aligned}
 P[\text{error}] &\leq \sum_{i=0}^m \binom{n}{i} \left(\frac{1}{2} + \delta\right)^i \left(\frac{1}{2} - \delta\right)^{n-i} \\
 &= \left(\frac{1}{2} + \delta\right)^m \left(\frac{1}{2} - \delta\right)^{m+1} \sum_{i=0}^m \binom{n}{i} \frac{\left(\frac{1}{2} - \delta\right)^{m-i}}{\left(\frac{1}{2} + \delta\right)^{m-i}} \\
 &\leq \left(\frac{1}{2} + \delta\right)^m \left(\frac{1}{2} - \delta\right)^{m+1} \sum_{i=0}^m \binom{n}{i}
 \end{aligned}$$

$$\sum_{i=0}^n \binom{n}{i} = 2^n = 2^{2m+1}$$

$$\sum_{i=0}^m \binom{n}{i} \geq \frac{1}{2} \times 2^{2m+1} = 2^{2m}$$

$$\begin{aligned}
 \text{Eq ① :- } P[\text{error}] &\leq \left(\frac{1}{2} + \delta\right)^m \left(\frac{1}{2} - \delta\right)^{m+1} 2^{2m} \\
 &= \left(\frac{1}{4} - \delta^2\right)^m \left(\frac{1}{2} - \delta\right) 2^{2m} \\
 &= (1 - 4\delta^2)^m \left(\frac{1}{2} - \delta\right) 2^{+ \delta} \\
 &\leq \frac{(1 - 4\delta^2)^m}{2}
 \end{aligned}$$

suppose we want to lower the error probability to some value γ . where $0 < \gamma < \frac{1}{2} - \delta$

We need to choose m so that

$$\frac{(1 - 4\delta^2)^m}{2} \leq \delta$$

$$m \log(1 - 4\delta^2) - 1 \leq \log \delta$$

$$m = \left\lceil \frac{1 + \log \delta}{\log(1 - 4\delta^2)} \right\rceil$$

Example :

suppose we start with a Monte Carlo Algorithm that gives correct answer with probability at least 0.55 so $\delta = 0.05$. If we desire a MC algorithm in which probab of error is at most 0.05, then it suffices to take $m = 230$, $n = 467$.

(k, l) - BBS Generator can be ϵ -distinguished from 1 random bits $\Rightarrow (\epsilon/l)$ previous bit prediction for (k, l) - BBS Generator $\Rightarrow$ Distinguishing algorithm for composite quadratic residues that is correct with probability at $\frac{1}{2} \pm \frac{\epsilon}{l}$

$\Rightarrow$ Unbiased Monte carlos algorithm for composite Quadratic Residues from at most $\frac{1}{2} \pm \frac{\epsilon}{l}$.

$\Rightarrow$ unbiased Monte-Carlo Algorithm for composite Quadratic Residues having error probability at most δ , for any $\delta > 0$.

Efficiency: The seq of PRB is contracted by taking the lsb of each s_i where $s_i = s_0^2 \bmod n$. Suppose instead we extract r , lsb bits of each seq for some +ve integer r but we need to ask if PRB. will remain secure (assuming intractability of composite quadratic residues). It has been shown that this approach is secure - provided $r \leq \log \log n$. so, we can extract $\log \log n$. pseudo random bits for modular sequencing.

In realistic implementation BBS generator, say one in which $n \approx 10^{160}$, we can extract 9 bits per squaring

Linear Feedback Shift Register (LFSR)

Random Number Generator (Loring Cipher) $\rightarrow$ Here speed is more important than security.

0 1 0 0 0 0 1 0 0 1 0 1 1

can be described by giving initial

values: $x_1 \equiv 0, x_2 \equiv 0, x_3 \equiv 0, x_4 \equiv 0, x_5 \equiv 0$.

Linear Recurrence Theorem

$$x_{n+5} \equiv x_n + x_{n+2} \pmod{2}$$

$$x_6 \equiv x_1 + x_3 \pmod{2}$$

$$\equiv 0 + 0$$

$$\equiv 0$$

$$x_7 \equiv x_2 + x_4 \pmod{2}$$

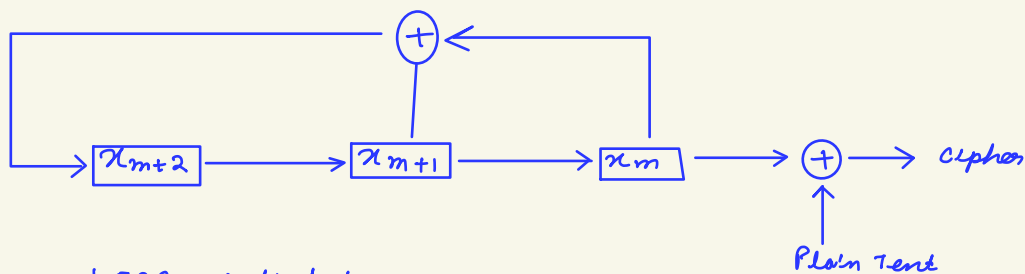
$$\equiv 1 + 0 \pmod{2} \equiv 1 \pmod{2}$$

More generally:-

$$x_{n+m} \equiv c_0 x_n + c_1 x_{n-1} + \dots + c_{m-1} x_{n-m+1} \pmod{2}$$

Encryption	{	Plaintext	1	0	1	1	0	0	1	
		Key	0	1	0	0	0	0	1	
		Ciphertext:	1	1	1	1	0	0	0	
			}							
			Decryption							
		$\frac{1}{2} +$ Key	:	0	1	0	0	0	0	1
			1	0	1	1	0	0	1	← Plain text

→ One advantage of LFSR is that a key with a large period can be generated using very little information.



LFSR satisfying

$$x_{m+3} = x_{m+1} + x_m$$

→ For each increment of a counter the bit in each box is shifted to other boxes as indicates with $\oplus$ denoting addition mod 2 of the incoming bit. The output which is bit x_m is added to the next bit of plain text to produce ciphertext. Here, LFSR represents the recurrence

$$x_{m+3} \equiv x_{m+1} + x_m \pmod{2}$$

Once the initial values of x_1 , x_2 and x_3 are specified the m/c produces subsequent bits very efficiently. Goal is to use the portion of the key to deduce the coefficients of the recurrence consequently compute the rest of the key.

Initial segment of keystream:

0 1 1 0 1 0 1 1 1 1 0 0

Start with length 2 recurrence

$$x_{n+2} = c_0 x_n + c_1 x_{n+1}$$

let $n=1$ and $n=2$

known values $x_1=0$, $x_2=1$, $x_3=1$, $x_4=0$

$$n=1$$

$$x_3 = c_0 x_1 + c_1 x_2$$

$$1 = c_0 \cdot 0 + c_1 \cdot 1 \quad \text{--- (1)}$$

$$n=2$$

$$x_4 = c_0 x_2 + c_1 x_3$$

$$0 = c_0 \cdot 1 + c_1 \cdot 1 \quad \text{--- (2)}$$

In matrix form

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{sol}^n: c_0 = c_1 = 1$$

$$x_{n+2} = x_n + x_{n+1}$$

Unfortunately:

$x_6 \neq x_4 + x_5$ - Therefore, we try with length 3 recurrence.

$$0 \mid 1 \mid 0 \mid 1 \mid 0 \mid 1 \mid 1 \mid 0 \mid 0$$

$$x_{n+3} = c_0 x_n + c_1 x_{n+1} + c_2 x_{n+2}$$

$$x_4 = 0 = c_0 \cdot 0 + c_1 \cdot 1 + c_2 \cdot 1$$

$$x_5 = 1 = c_0 \cdot 1 + c_1 \cdot 1 + c_2 \cdot 0$$

$$x_6 = 0 = c_0 \cdot 1 + c_1 \cdot 0 + c_2 \cdot 1$$

In matrix form:

$$\underbrace{\begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}}_{\det^m = 0} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Consider length 4 recurrence

The matrix eqⁿ is

$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

Solⁿ: $C_0 = 1, C_1 = 1, C_2 = 0, C_3 = 0$

Resulting Recurrence

$$x_{n+4} = x_n + x_{n+1} \pmod{2}$$

In general to test for a recurrence of length m
we assume $x_1, x_2, \dots, x_m$

$$\begin{pmatrix} x_1 & x_2 & \dots & x_m \\ x_2 & x_3 & \dots & x_{m+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_m & x_{m+1} & \dots & x_{m+m} \end{pmatrix} \begin{pmatrix} C_0 \\ C_1 \\ \vdots \\ C_{m-1} \end{pmatrix} = \begin{pmatrix} x_m \\ x_{m+1} \\ \vdots \\ x_{2m} \end{pmatrix}$$

LF SR

A strategy for finding the coefficient of the recurrence:

Suppose we know int 100 bits of the key. For $m = 2, 3, 4$ form $m \times m$ matrix and compute its determinant.

If several consecutive values of m yield 0 determinants, stop.
 The last m to yield non-zero determinant ($1 \pmod 2$) is probably the length of the recurrence.

eg - seq of bits

1 0 0 1 1 0 0 1 0 0 1 1 1 0 0 0 1 1 0 0 0 1 0 1 0 0 0 ...

The first 20 determinants

1, 0, 1, 0, 0, 1, 0, 1, 0, 0, ----- 0
 12 0's

A reasonable guess $m=8$

$$\{c_0, c_1, c_2, \dots, c_7\} = \{1, 1, 0, 0, 1, 0, 0, 0\}$$

Guessed recurrence:

$$x_{n+8} = x_n + x_{n+1} + x_{n+4}$$

Suppose we know bits from the middle of the seq.

$$x_{17} = 1, \quad x_{18} = 0, \quad x_{19} = 0$$

Write the recurrence as:

$$x_n \equiv x_{n+1} + x_{n+4} + x_{n+8}$$

We are working mod 2 $\Rightarrow$

$$-x_n = x_n \quad \& \quad -x_{n+8} = x_{n+8}$$

$$\therefore x_{16} = x_{17} + x_{20} + x_{24}$$

$$= 1 + 0 + 1$$

$$= 0$$

Continuing in this way we get

$$x_{15}, x_{14}, \dots, x_1$$

Proposition: Let $x_1, x_2, x_3, \dots$ be a seq of bits produced by a linear recurrence mod 2. For each $n \geq 1$ let

$$M_n = \begin{pmatrix} x_1 & x_2 & \dots & x_n \\ x_2 & x_3 & \dots & x_{n+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_n & x_{n+1} & \dots & x_{2n-1} \end{pmatrix}$$

Let N be the length of the shortest recurrence that generates the seq $x_1, x_2, x_3, \dots$. Then

$$\det(M_N) \equiv 1 \pmod{2} \text{ and}$$

$$\det(M_n) \equiv 0 \pmod{2} \text{ for}$$

$$\text{all } n > N$$

Pf: Note
length 3 recurrence

$$x_{n+3} \equiv x_{n+2}$$

(shorter relation)

$$x_{n+1} \equiv x_n \quad (\forall n \geq 2)$$

$$\text{seq: } 1, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1$$

Recurrence: $x_{n+4} \equiv x_{n+3} + x_{n+1} + x_n$

$$x_4 \quad x_5 = 1 + 0 + 1 = 0$$

A shorter recurrence

$$x_{n+2} \equiv x_{n+1} + x_n$$

• If there is a recurrence of length N & if $n > N$ then one row of matrix M_n is congruent mod 2 to a linear combination of other rows.

For example in this case $x_{n+2} \equiv x_{n+1} + x_n$ fourth row is the sum of 1st and 3rd rows $\det(M_n) \equiv 0 \pmod{2} \quad \forall n > N$
suppose $\det(M_n) \equiv 0 \pmod{2} \Rightarrow \exists$ a non-zero vector

$\vec{b} = (b_0, b_1, \dots, b_{n-1})$ such that $\vec{b} \cdot M_n \equiv 0$. we could

show that this gives a recurrence relation for the seq

$x_1, x_2, x_3, \dots$ & that the length of this relation

is less than N . This contradicts the assumption

that N is smallest

The contradiction implies $\det(M_n) \equiv 1 \pmod{2}$.

Let the recurrence of length N be:

$$x_{n+N} \equiv c_0 x_n + c_1 x_{n+1} + \dots + c_{N-1} x_{n+N-1}$$

For each $i \geq 0$, let

$$M^{(i)} = \begin{pmatrix} x_{i+1} & x_{i+2} & \dots & x_{i+N} \\ x_{i+2} & x_{i+3} & \dots & x_{i+N+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_{i+N} & x_{i+N+1} & \dots & x_{i+2N-1} \end{pmatrix}$$

Then $M^{(0)} = M_N$. The recurrence relation implies that

$$M^{(i)} \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{pmatrix} = \begin{pmatrix} x_{i+n+1} \\ x_{i+n+2} \\ \vdots \\ x_{i+2N} \end{pmatrix}$$

which is the last column of $M^{(i+1)}$

By the choice of B we have $B \cdot M^{(0)} = \bar{0}$. $M_N = 0$. Suppose that we know $B \cdot M^{(i)} = 0$, for some i . Then,

$$\bar{0} \begin{pmatrix} x_{i+n+1} \\ x_{i+n+2} \\ \vdots \\ x_{i+2N} \end{pmatrix} = B \cdot M^{(i)} \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{pmatrix} \equiv 0 \pmod{i}$$

Therefore B annihilates the last column of $M^{(i+1)}$. Since, the remaining columns of $M^{(i+1)}$ are columns of $M^{(i)}$ are columns of $M^{(i)}$, we find that $B \cdot M^{(i+1)} \equiv 0$. By

induction, we obtain $B \cdot M^{(i)} \equiv 0 \quad \forall i \geq 0$. Let $n \geq 1$.

The first column of M yields,

$$b_0 x_n + b_1 x_{n+1} + \dots + b_{n-1} x_{n+N+1} \equiv 0$$

Since $\vec{b}$ is not a zero vector $b_j \neq 0$ for at least one j . Let m be the length of $\vec{b}$, such that $b_j \neq 0$, which means $b_m = 1$.

$$b_m x_{n+m-1} = -x_{n+m-1}$$

$\therefore$ we rearrange the relation,

$$x_{n+m-1} = b_0 x_n + b_1 x_{n+1} + \dots + b_{m-1} x_{n+m-2}$$

This is a recurrence of length $m-1$, since, $m-1 < N$ & N is assumed to be the shortest possible length we have a contradiction.

$\therefore$ The answer is $\det(M_N) = 0$ must be and so $\det(M_N) \equiv 1 \square$

LFSR

$\rightarrow$ Suppose the length of the recurrence is m . Any m consecutive terms of the sequence determine all future elements and by reversing the recurrence all previous values too.

$\rightarrow$ Clearly if we have m consecutive 0's then all future and previous values are 0's. Therefore, we exclude the case from consideration. There are $2^m - 1$ strings of 0's and 1's of length m in which at least one term is non-zero.

$\therefore$ As soon as there are more than $2^m - 1$ terms, some strings of length m , must occur twice so the sequence repeats. The period of the sequence is at most

$$2^m - 1.$$

Associated to the recurrence

$$x_{n+m} \equiv c_0 x_n + c_1 x_{n-1} + \dots + c_{m-1} x_{n+m-1} \pmod{2}$$

there is a polynomial

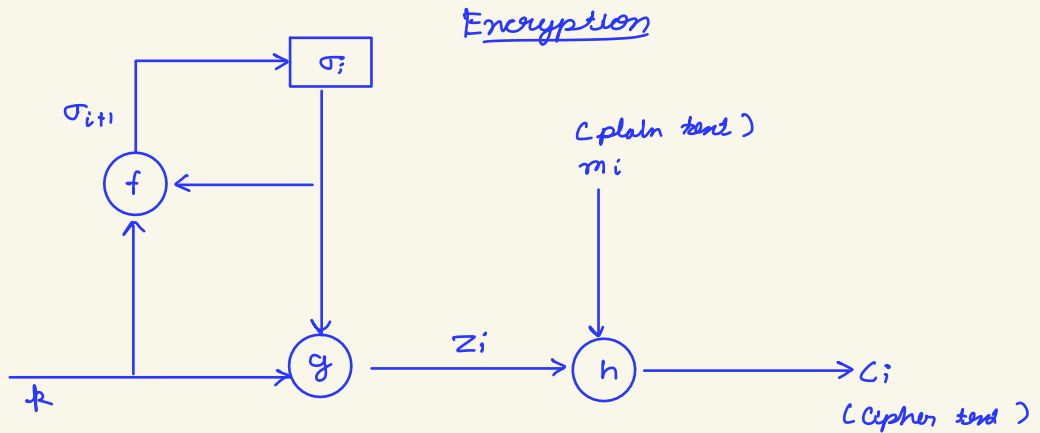
$$f(T) = T^m - c_{m-1}T^{m-1} - \dots - c_0$$

If $f(T)$ is irreducible (cannot factorise), then it can be shown that the period divides $2^m - 1$. An interesting case when $2^m - 1$ is prime (Mersenne Prime) the period is maximal, namely $2^m - 1$.

Stream Ciphers

1) Synchronous stream cipher:

Defⁿ:- Here keystream is generated independently of plaintext and cipher text.



key $\in K$

Keystream: Z_i

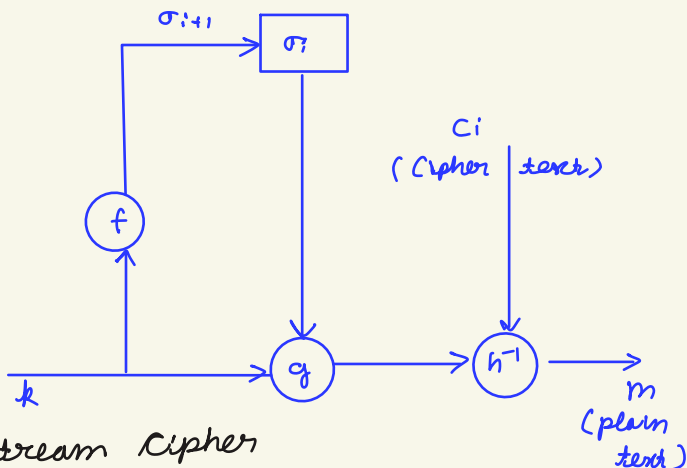
State: σ_i

$$\sigma_{i+1} = f(\sigma_i, k)$$

$$Z_i = g(\sigma_i, k)$$

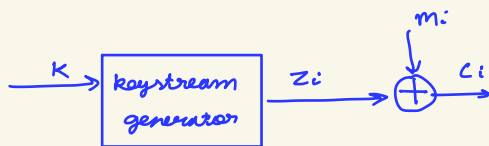
$$C_i = h(Z_i, m_i)$$

Decryption

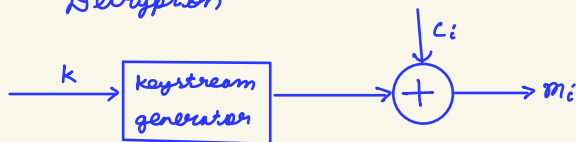


2 Binary Additive stream cipher

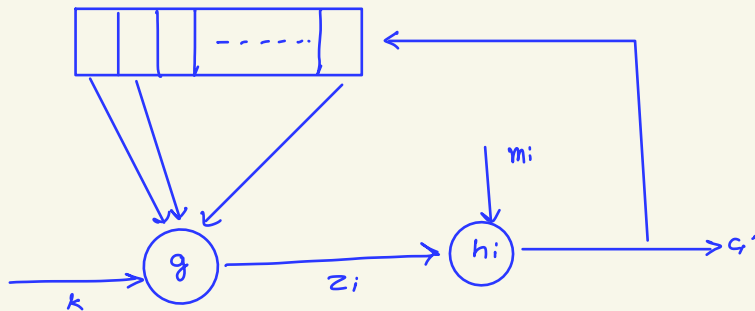
Encryption



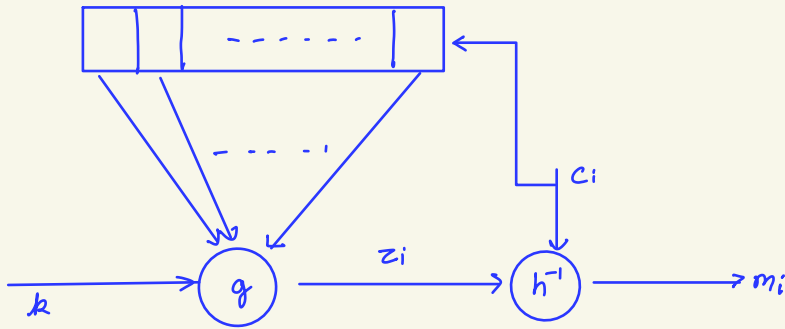
Decryption



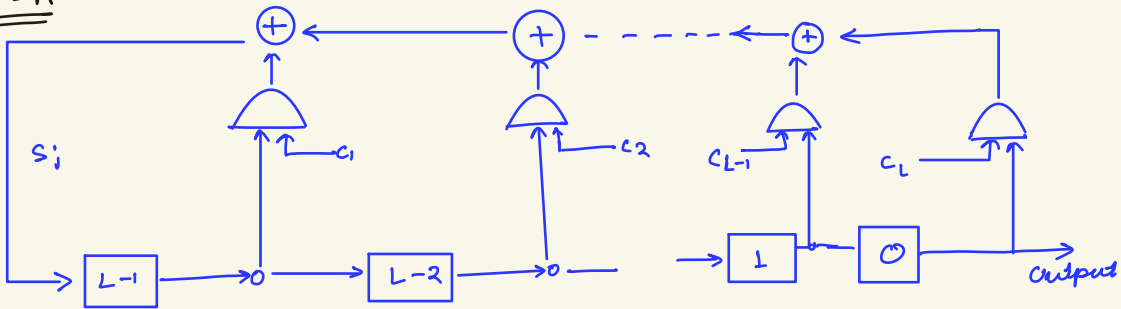
Self synchronizing stream cipher



Decryption



LFSR



LFSR of length L $\langle L, C(D) \rangle$ where

$$C(D) = 1 + c_1 D + c_2 D^2 + \dots + c_L D^L$$

Connection
polynomial

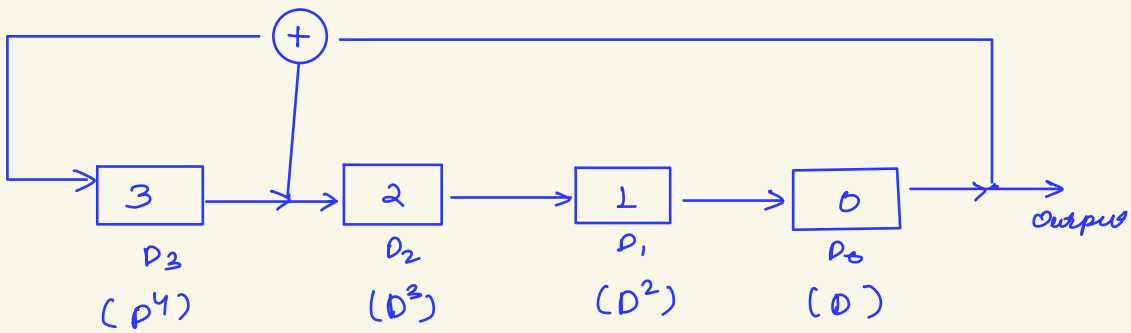
→ LFSR is said to be non-singular if degree of $C(D)$ is 2 (i.e., $c_L = 1$)

→ Initial state of LFSR is $[s_{L-1}, \dots, s_2, s_1, s_0]$, then the output sequence:

$$S = c_0, s_1, c_2, \dots$$

where: $s_j = (c_1 s_{j-1} + c_2 s_{j-2} + \dots + c_L s_{j-L}) \bmod 2$

$$\forall j > L$$



$$C(D) = 1 + D + D^4$$

$$\text{LFSR} = \langle 4, 1 + D + D^4 \rangle$$

Output seq. (D_0)

0, 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 1, 0, 1, - - - -

It is periodic with period 15.

Fact: Every output seq. of LFSR $\langle L, C(D) \rangle$ is periodic if and only if the connection polynomial $C(D)$ has degree L .

Fact: Let $C(D) \in \mathbb{Z}_2(D)$ be a connection polynomial of degree L .

(i) If $C(D)$ is irreducible over $\mathbb{Z}_2$ then each of $2^L - 1$ non-zero initial state of the non-negative LFSR $\langle L, C(D) \rangle$ produces an output sequence with period equal to the least positive integer N such that $C(D)$ divides $1 + D^N$.

LFSR $[N$ is a divisor of $2^L - 1]$

(ii) If $C(D)$ is a primitive polynomial then each of the $2^L - 1$ non-zero initial states of non-singular LFSR $\langle L, C(D) \rangle$ produces an output seq of the maximum possible period $2^L - 1$.

LFSR

Defⁿ:- If $C(D) \in \mathbb{Z}_2[D]$ is a primitive polynomial of degree L then $\langle L, C(D) \rangle$ is called maximum length LFSR. The output of maximum length LFSR with non-zero initial state is called m-sequence.

FACT:- Let S be a m-sequence that is generated by maximum length LFSR of length L

(i) Let k be an integer, $1 \leq k \leq L$ and $\bar{s}$ be any subsequence of S of length $2^{L-k} - 1$. Then each non-zero sequence of length k appears exactly 2^{L-k} times as a subsequence of $\bar{s}$. Furthermore, the 0-seq of length k appears exactly $2^{L-k} - 1$ times as a subsequence of $\bar{s}$. In other words, the distribution of patterns having fixed length of almost L is almost uniform.

(ii) S satisfies Golomb's randomness postulates, i.e., every m-sequence is also pn-sequence.

Linear complexity

Defⁿ: The linear complexity of an infinite binary sequence s denoted by $L(s)$, is defined as follows:-

- (i) If s is the 0 seq, $s = 0, 0, \dots, 0$ then $L(s) = 0$.
- (ii) If no LFSR generates s , then $L(s) = \infty$.
- (iii) Otherwise $L(s)$ is the length of the shortest LFSR that generates s .

Properties

Let s and t be binary sequences.

- (i) For any $n > 1$, the linear complexity of any subsequence s^n satisfies $0 \leq L(s^n) \leq n$

(ii) $L(s^n) = 0$ if and only if s^n is the 0-seq of length n .

(iii) $L(s^n) = n$ iff

$$s^n = 0, 0, 0, \dots, 1 \rightarrow \text{last bit } n$$

(iv) If s is periodic with period N , then $L(s) \leq N$.

(v) $L(s \oplus t) \leq L(s) + L(t)$

where, $s \oplus t$ denotes bitwise XOR of s and t .

Fact:- If $C(D) \in \mathbb{Z}_2[D]$

is irreducible over $\mathbb{Z}_2$ and has degree L , then each of the

$2^L - 1$, non-zero initial states of the non-singular

LFSR $\langle L, C(D) \rangle$ produces an output sequence with linear

complexity L .

Linear complexity of Random sequence

$S = s_0, s_1, \dots$, be a binary sequence and L_N denote the linear complexity of the subsequence

$$S^N = s_0, s_1, \dots, s_{N-1} \quad N \geq 0$$

The sequence $L_1, L_2, \dots$ is called the linear complexity profile of S .

Fact: Let $L_1, L_2, \dots$ be the linear complexity profile of a sequence $S_0 = s_0, s_1, \dots$

(i) If $j > i$ then $L_j \geq L_i$

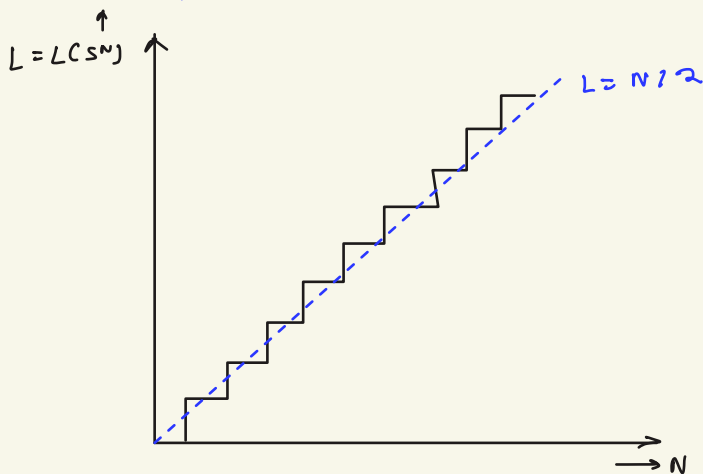
(ii) $L_{N+1} > L_N$ is possible iff $L_N \leq N/2$

(iii) $L_{N+1} > L_N$, then $L_{N+1} + L_N = N + 1$.

Example $S^{20} = 1, 0, 0, 1, 0, 0, 1, 1, 1, \dots$

L_1, L_2, L_3, L_4

$$L(S^N) = 1, 2, 2, 3, 3, 3, 3, 5, 5, 5, 6, 6, 6, 8, 8, 8, 9$$



For a Random sequence :-

$$\text{If } L(S^N) = \left\lfloor \frac{N+1}{2} \right\rfloor \quad \forall N \geq 1$$

(Necessary but not sufficient)

Ex:-

$$s_i = \begin{cases} 1 & \text{if } i = 2^j - 1 \text{ for some } j \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

↓
follows $L = \frac{N}{2}$ line but non-random

Berlekamp Massey Algorithm

→ To compute linear complexity profile.

Defⁿ:- Consider the finite binary sequence

$$S^{N+1} = s_0, s_1, \dots, s_N$$

For $C(D) = 1 + C_1 D + \dots + C_L D^L$, let $\langle L, C(D) \rangle$ be an LFSR that generates the sequence $S^N = s_0, s_1, \dots, s_{N-1}$. The next discrepancy L_N is the difference between s_N and $(N+1)^{\text{st}}$ term generated by LFSR.

$$L_N = (s_N + \sum_{i=1}^L C_i s_{N-i}) \bmod 2$$

• Fact:- Let $S^N = s_0, s_1, \dots, s_{N-1}$ be a finite binary seq of linear complexity.

$L = L(S^N)$ and let $\langle L, C(D) \rangle$ be a LFSR which generates S^N

(i) LFSR $\langle L, C(0) \rangle$ also generates $S^{N+1} = s_0, s_1, \dots, s_{N-1}, s_N$ iff next discrepancy L_N equals 0.

(ii) If $L_N = 0$, then $L(S^{N+1}) = L$

(iii) If $L_N = 1$, let m be the largest integer $< N$, such that $L(S^m) < L(S^N)$ and let $\langle L(S^m), B(0) \rangle$ be an LFSR of length $L(S^m)$ which generates s^m . Then $\langle L', C'(0) \rangle$ is an LFSR of smallest length which generates S^{N+1} , where

$$L' = \begin{cases} L & \text{if } L > N/2 \\ N+1-L & \text{if } L \leq N/2 \end{cases}$$

$$\text{and } C'(0) = C(0) + B(0) 0^{N-m}$$

INPUT: A binary seqⁿ $s^n = s_0, s_1, \dots, s_{N-1}$ of length n .

OUTPUT: The linear complexity $L(s^n)$ of s^n , $0 \leq L(s^n) \leq n$.

1) INITIALIZATION: $C(0) \leftarrow 1$, $L \leftarrow 0$, $m \leftarrow -1$, $B(0) \leftarrow 1$, $N \leftarrow 0$.

2) While $(N < n)$ do the following:

2.1) Compute the next discrepancy d

$$d \leftarrow (s_N + \sum_{i=1}^L c_i s_{N-i}) \bmod 2$$

2.2) If $d = 1$, then do the following:

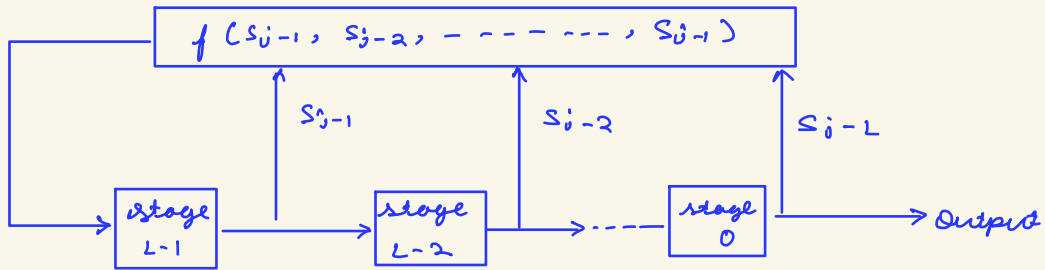
$$T(0) \leftarrow C(0), \quad C(0) \leftarrow C(0) + B(0) 0^{N-m}$$

if $(L \leq N/2)$ then,

$$2.3) \quad N \leftarrow N+1, \quad L \leftarrow N+1-L, \quad m \leftarrow N, \quad B(0) \leftarrow T(0)$$

3) return (L)

Non-linear - LFSR



$$S_j = f(S_{j-1}, S_{j-2}, \dots, S_{j-L})$$

Defⁿ :- If the period of output sequence (for any initial steps) of a non-singular FSR of length L , is 2^L , then FSR is called a de Bruijn FSR and output sequence is called de Bruijn sequence.

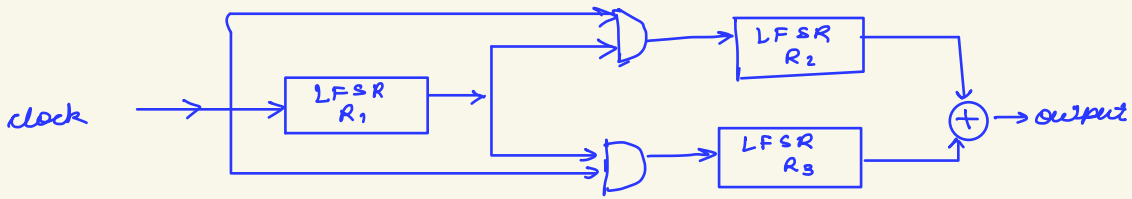
Example : $f(x_1, x_2, x_3) = 1 \oplus x_2 \oplus x_3 \oplus x_1 x_2$
for a 3-stage FSR

i	stage 2	stage 1	stage 0
0	0	0	0
1	1	0	0
2	1	1	0
3	1	1	1
4	0	1	1

Output seq $\rightarrow$ 0, 0, 0, 1, 1, 1, 0, 1

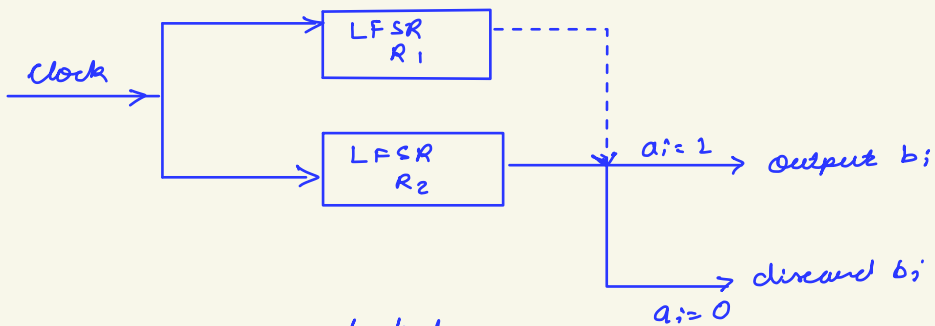
Clock Controlled Generator :-

1) Alternating step Generator :-



$$\text{Period} : 2^{L_1}(2^{L_2}-1)(2^{L_3}-1)$$

2) Shrinking Generator :-



1) Registers R_1 and R_2 are clocked.

2) If R_1 is 1, the output bit of R_2 forms part of the keystream.

3) If the output R_1 is 0, the output bit of R_2 is discarded.

Fact :

If R_1 and R_2 are maximum LFSR L_1, L_2 respectively and let π be the output seq of shrinking generator then.

(i) If $\gcd(L_1, L_2) = 1$, then π has period $(2^{L_2}-1)(2^{L_1}-1)$.